

Portfolio Milestone #1: Bank Marketing Dataset Exploration (SAS Studio)

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Introduction

Beginning with the Bank Marketing Dataset (bank.csv) and SAS for the summary stats and graphics for the variables. The variables include each customer's demographic data, the details of the marketing campaign for each customer, and relevant economic data. The output variable is y (yes/no), which answers whether or not the customer has subscribed to a given term deposit. This is a first pass exploration of the data to help understand the structure of the data and the patterns and distributions to support classification modelling for the final portfolio project (Colorado State University Global, n.d.).

Dataset Upload and Structure SAS (Studio)

The dataset has been successfully uploaded and stored in SAS using the path: WORK.BANK. The dataset has 4,119 records and 21 variables in total, with 21 variables of mixed types, both categorical variables (job, marital, education, housing, loan, contact, month, poutcome, y) and numerical variables (age, duration, campaign, pdays, previous, euribor3m, emp.var.rate, cons.price.idx, cons.conf.idx, nr.employed). This is an important factor at this stage as each of the numerical variables will require summary statistics, and histograms, and each of the categorical variables will require frequency tables and bar charts.

Summary Statistics (Numeric Variables)

Summary statistics for numeric variables are generated using PROC MEANS. Several characteristics of the output are quite noteworthy.

- Age shows typical working-age distribution with values around the middle comprising most of the clients. Since age is not too skewed, it can be seen that mean and median are close to each other.]

- Duration shows large range and high skewness. Since the mean is larger than the median, it can be inferred that the distribution is right-skewed, meaning most calls are short in duration, and that there is a small number of very long calls.
- The variables related to the campaign, namely campaign, pdays, and previous, suggest that many customers have been contacted a small number of times, while a small group of customers have much larger values (potential outliers).
- The economic variables (euribor3m, emp.var.rate, cons.price.idx, cons.conf.idx, nr.employed) are indicators of the market and can be associated with the decision related to subscription.

The summary statistics indicate that the dataset contains a mixture of stable variables (age) and variables such as duration and campaign metrics, which are variable/skewed and may require special consideration for future modeling.

Graphical Summaries

Histogram of Numeric Variables

Histogram is a preferred method to show the distribution of numeric variables. The histogram for age shows that the majority of customers are in a middle age range which supports the summary statistics that age is not very skewed. The duration histogram, however, has a strong right skew, which means most calls are short, but there are a few exceptionally long calls. Such outliers may negatively impact the performance of the model and may need special attention in the analysis.

Bar Charts and Frequencies for Categorical Variables

PROC FREQ and bar charts assisted in summarizing categorical variables. In the frequency tables, several categories overshadow the others (for instance, some job or education

levels are more frequent). An important finding is how the target variable y is spread out; this highlights the imbalance present in the data set. The overwhelming majority of the observations record a subscription status of “no,” with a significantly smaller number recording “yes.” This is a critical imbalance since it may give the classification models the false appearance of being accurate, interrelating to the majority class. Therefore, the assessment of the models will need to encompass more than just accuracy.

Appropriate Classification Technique

As the response variable y is binary, (yes/no), the valid data mining technique to use for this dataset is classification. A baseline classification model applicable to this dataset is logistic regression, as it predicts the probability of the binary outcome and has interpretations that can be utilized in marketing. It can predict which characteristics of the customer and the campaign will increase the possibility of a subscription. Once more data is available in the later stages of the portfolio, additional classification techniques (like decision trees or random forests) may be used, in the event that additional complexity is present.

Key Data Aspects Identified

From the summary statistics and the various graphs, the following data aspects were deduced:

- The dataset has both categorical and numeric variables, which means that it requires separate techniques for summarization.
- Variables like duration and campaign measures that show extreme values (right skew) and outliers like duration and campaign measures may impact model training.
- Some categorical variables contain the word “unknown”. This can be seen as incomplete data, which can impact the outcome of the predictions.

- The imbalance of classes in y : There are more “no” outcomes than “yes” outcomes, and this should be taken into consideration in the classification evaluation.

This milestone lays the groundwork for the Module 8 portfolio project, where future tasks include the construction of models for anomaly detection, association analyses, clustering of numeric attributes, and classification.

References

Colorado State University Global. (n.d.). Writing Center: APA resources and templates.

APPENDIX A — Screenshots

Bank Marketing Dataset Structure (WORK.BANK) System Date: 11JAN2026 The CONTENTS Procedure					
Data Set Name	WORK.BANK			Observations	4119
Member Type	DATA			Variables	21
Engine	V9			Indexes	0
Created	01/11/2026 12:25:01			Observation Length	176
Last Modified	01/11/2026 12:25:01			Deleted Observations	0
Protection				Compressed	NO
Data Set Type				Sorted	NO
Label					
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64				
Encoding	utf-8 Unicode (UTF-8)				

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	6
First Data Page	1
Max Obs per Page	743
Obs in First Data Page	718
Number of Data Set Repairs	0
Filename	/saswork/SAS_workBB6F0001743E_odaws01-usw2-2.oda.sas.com/SAS_work183E0001743E_odaws01-usw2-2.oda.sas.com/bank.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	402653219
Access Permission	rw-r--r--
Owner Name	u64301103
File Size	896KB
File Size (bytes)	917504

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	age	Num	8	BEST12.	BEST32.
12	campaign	Num	8	BEST12.	BEST32.
18	cons.conf.idx	Num	8	BEST12.	BEST32.
17	cons.price.idx	Num	8	BEST12.	BEST32.
8	contact	Char	9	\$9.	\$9.
10	day_of_week	Char	3	\$3.	\$3.
5	default	Char	7	\$7.	\$7.
11	duration	Num	8	BEST12.	BEST32.
4	education	Char	19	\$19.	\$19.
16	emp.var.rate	Num	8	BEST12.	BEST32.
19	euribor3m	Num	8	BEST12.	BEST32.
6	housing	Char	7	\$7.	\$7.
2	job	Char	13	\$13.	\$13.
7	loan	Char	7	\$7.	\$7.
3	marital	Char	8	\$8.	\$8.
9	month	Char	3	\$3.	\$3.
20	nr.employed	Num	8	BEST12.	BEST32.
13	pdays	Num	8	BEST12.	BEST32.
15	poutcome	Char	11	\$11.	\$11.
14	previous	Num	8	BEST12.	BEST32.
21	y	Char	3	\$3.	\$3.

Generated in SAS Studio on 11JAN2026:12:25:01.

Table 1.

WORK Library Dataset List Showing WORK.BANK

Bank Marketing Dataset Structure (WORK.BANK) System Date: 11JAN2026															
Obs	age	job	marital	education	default	housing	loan	contact	month	day_of_week	duration	campaign	pdays	previous	pot
1	30	blue-collar	married	basic.9y	no	yes	no	cellular	may	fri	487	2	999	0	non
2	39	services	single	high.school	no	no	no	telephone	may	fri	346	4	999	0	non
3	25	services	married	high.school	no	yes	no	telephone	jun	wed	227	1	999	0	non
4	38	services	married	basic.9y	no	unknown	unknown	telephone	jun	fri	17	3	999	0	non
5	47	admin.	married	university.degree	no	yes	no	cellular	nov	mon	58	1	999	0	non
6	32	services	single	university.degree	no	no	no	cellular	sep	thu	128	3	999	2	failu
7	32	admin.	single	university.degree	no	yes	no	cellular	sep	mon	290	4	999	0	non
8	41	entrepreneur	married	university.degree	unknown	yes	no	cellular	nov	mon	44	2	999	0	non
9	31	services	divorced	professional.course	no	no	no	cellular	nov	tue	68	1	999	1	failu
10	35	blue-collar	married	basic.9y	unknown	no	no	telephone	may	thu	170	1	999	0	non

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Table 2. PROC CONTENTS Output for Bank Marketing Dataset (WORK.BANK).

Summary Statistics for Numeric Variables (WORK.BANK) System Date: 11JAN2026 The MEANS Procedure						
Variable	N	Mean	Median	Std Dev	Minimum	Maximum
age	4119	40.1136198	38.0000000	10.3133615	18.0000000	88.0000000
duration	4119	256.7880554	181.0000000	254.7037361	0	3643.00
campaign	4119	2.5372663	2.0000000	2.5681592	1.0000000	35.0000000
pdays	4119	960.4221899	999.0000000	191.9227858	0	999.0000000
previous	4119	0.1903375	0	0.5417883	0	6.0000000
euribor3m	4119	3.6213557	4.8570000	1.7335912	0.6350000	5.0450000
emp.var.rate	4119	0.0849721	1.1000000	1.5631145	-3.4000000	1.4000000
cons.price.idx	4119	93.5797043	93.7490000	0.5793488	92.2010000	94.7670000
cons.conf.idx	4119	-40.4991017	-41.8000000	4.5945775	-50.8000000	-26.9000000
nr.employed	4119	5166.48	5191.00	73.6679036	4963.60	5228.10

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Table 3. PROC MEANS Summary Statistics for Numeric Variables.

Frequency Tables for Categorical Variables (WORK.BANK)

System Date: 11JAN2026

The FREQ Procedure

job	Frequency	Percent
admin.	1012	24.57
blue-collar	884	21.46
entrepreneur	148	3.59
housemaid	110	2.67
management	324	7.87
retired	166	4.03
self-employed	159	3.86
services	393	9.54
student	82	1.99
technician	691	16.78
unemployed	111	2.69
unknown	39	0.95

marital	Frequency	Percent
divorced	446	10.83
married	2509	60.91
single	1153	27.99
unknown	11	0.27

education	Frequency	Percent
basic.4y	429	10.42
basic.6y	228	5.54
basic.9y	574	13.94
high.school	921	22.36
illiterate	1	0.02
professional.course	535	12.99
university.degree	1264	30.69
unknown	167	4.05

default	Frequency	Percent
no	3315	80.48
unknown	803	19.50
yes	1	0.02

housing	Frequency	Percent
no	1839	44.65
unknown	105	2.55
yes	2175	52.80

contact	Frequency	Percent
cellular	2652	64.38
telephone	1467	35.62

month	Frequency	Percent
apr	215	5.22
aug	636	15.44
dec	22	0.53
jul	711	17.26
jun	530	12.87
mar	48	1.17
may	1378	33.45
nov	446	10.83
oct	69	1.68
sep	64	1.55

day_of_week	Frequency	Percent
fri	768	18.65
mon	855	20.76
thu	860	20.88
tue	841	20.42
wed	795	19.30

poutcome	Frequency	Percent
failure	454	11.02
nonexistent	3523	85.53
success	142	3.45

y	Frequency	Percent
no	3668	89.05
yes	451	10.95

Generated in SAS Studio on 11JAN2026:12:26:48.

Table 4. PROC FREQ Output for Categorical Variables

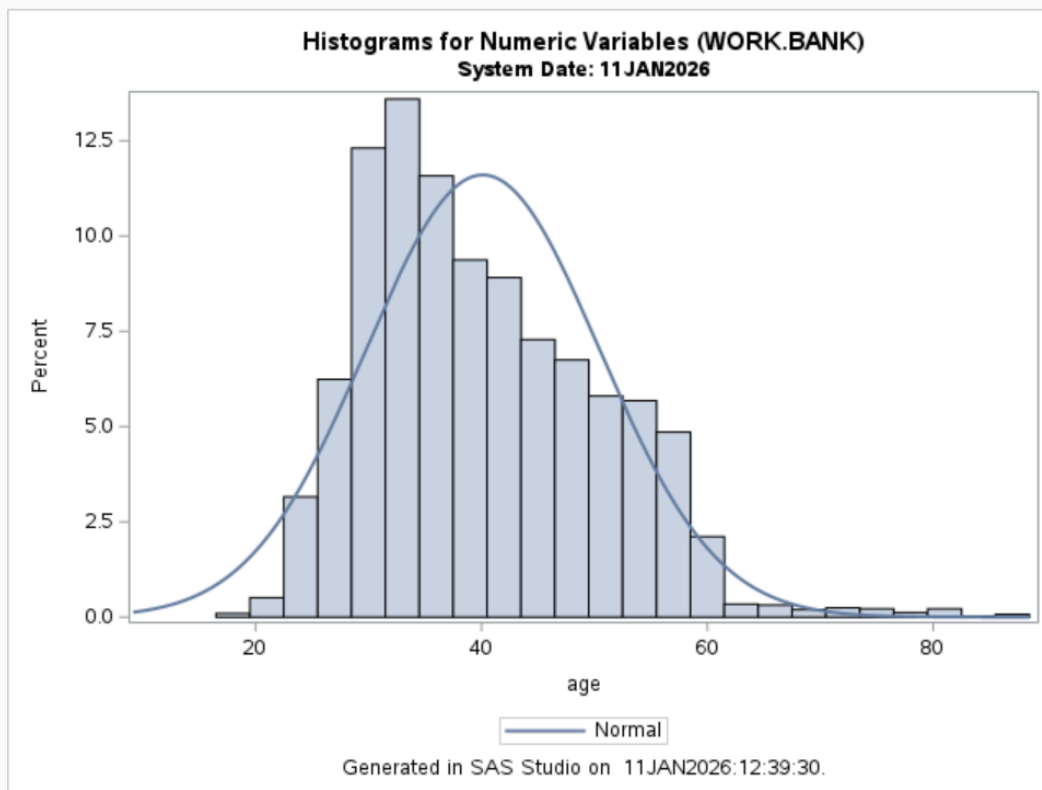


Figure 1. Histogram of Age.

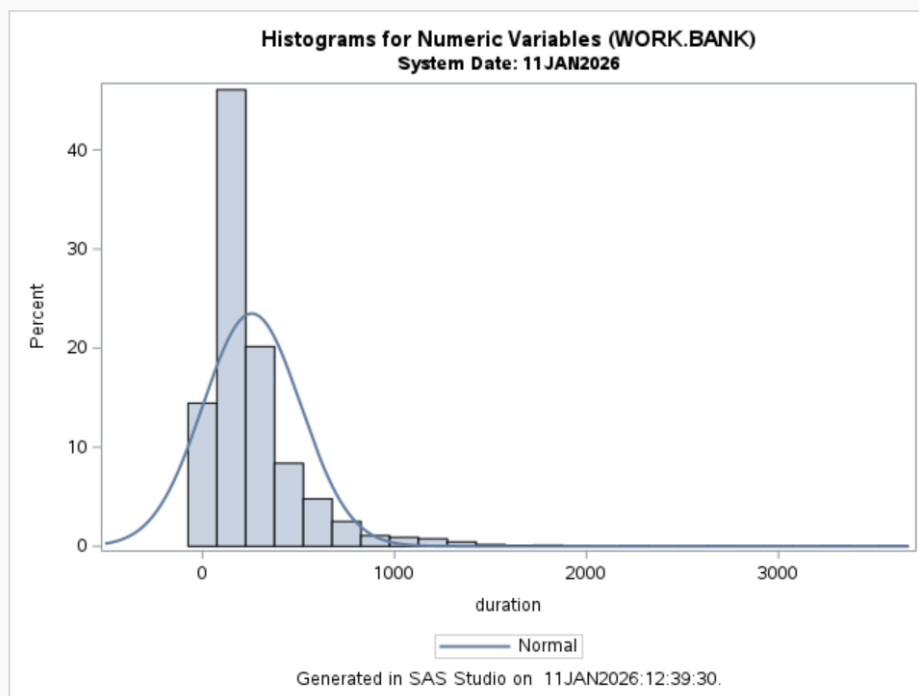


Figure 2. Histogram of Duration.

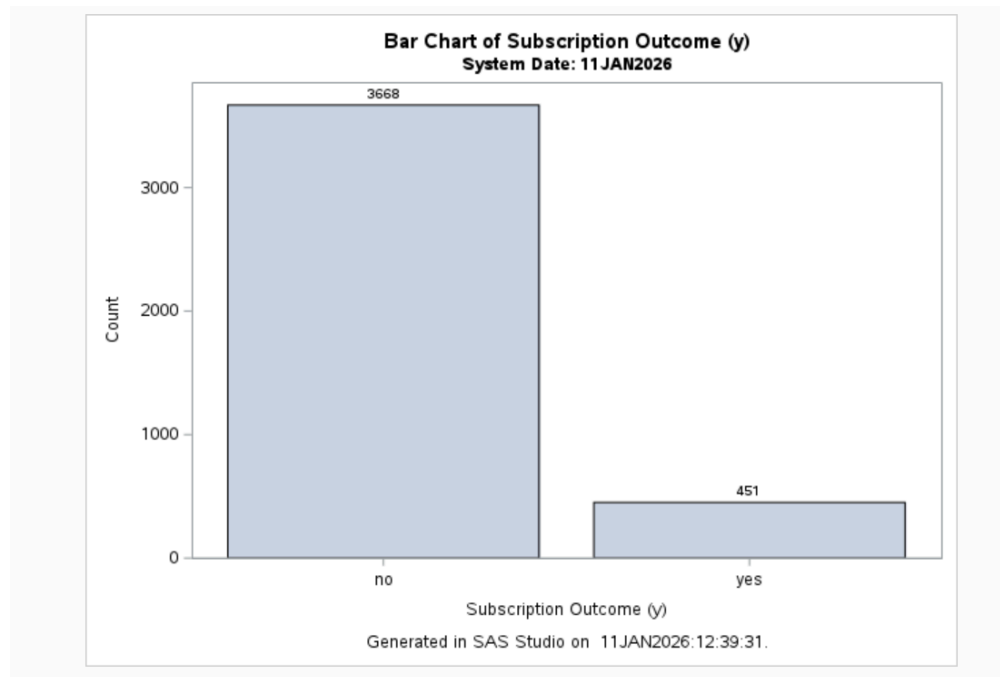


Figure 3. Bar Chart of Subscription Outcome (y).

Appendix B- Code Required

```
ods graphics on;

/* Always print timestamp on every output */
title1 "Bank Marketing Dataset (WORK.BANK)";
title2 "System Date: %sysfunc(today(), date9.)";
footnote1 "Generated in SAS Studio on %sysfunc(datetime(), datetime19.)";

/* 1) Dataset structure + first 10 rows (categorical + numeric shown) */
proc contents data=work.bank; run;

proc print data=work.bank(obs=10);
run;

/* 2) Summary statistics for ALL numeric variables */
title1 "Summary Statistics for Numeric Variables (WORK.BANK)";
title2 "System Date: %sysfunc(today(), date9.)";
footnote1 "Generated in SAS Studio on %sysfunc(datetime(), datetime19.)";

proc means data=work.bank n mean median std min max;
    var age duration campaign pdays previous euribor3m
        'emp.var.rate'n 'cons.price.idx'n 'cons.conf.idx'n 'nr.employed'n;
run;

/* 3) Frequency tables for ALL categorical variables */
title1 "Frequency Tables for Categorical Variables (WORK.BANK)";
title2 "System Date: %sysfunc(today(), date9.)";
footnote1 "Generated in SAS Studio on %sysfunc(datetime(), datetime19.)";
```

```

proc freq data=work.bank;

    tables job marital education default housing loan contact month

        day_of_week poutcome y / nocum;

run;


/* 4) Histograms for ALL numeric variables */

title1 "Histograms for Numeric Variables (WORK.BANK)";
title2 "System Date: %sysfunc(today(), date9.)";
footnote1 "Generated in SAS Studio on %sysfunc(datetime(), datetime19.).";


proc sgplot data=work.bank;

    histogram age; density age;

run;


proc sgplot data=work.bank;

    histogram duration; density duration;

run;


proc sgplot data=work.bank;

    histogram campaign; density campaign;

run;


proc sgplot data=work.bank;

    histogram pdays; density pdays;

run;


proc sgplot data=work.bank;

    histogram previous; density previous;

```

```
run;
```

```
proc sgplot data=work.bank;
```

```
    histogram euribor3m; density euribor3m;
```

```
run;
```

```
proc sgplot data=work.bank;
```

```
    histogram 'emp.var.rate'n; density 'emp.var.rate'n;
```

```
run;
```

```
proc sgplot data=work.bank;
```

```
    histogram 'cons.price.idx'n; density 'cons.price.idx'n;
```

```
run;
```

```
proc sgplot data=work.bank;
```

```
    histogram 'cons.conf.idx'n; density 'cons.conf.idx'n;
```

```
run;
```

```
proc sgplot data=work.bank;
```

```
    histogram 'nr.employed'n; density 'nr.employed'n;
```

```
run;
```

```
title; footnote;
```

```
ods graphics on;
```

```
title1 "Bar Chart of Subscription Outcome (y)";
```

```
title2 "System Date: %sysfunc(today()), date9.";
```

```
footnote1 "Generated in SAS Studio on %sysfunc(datetime()), datetime19..";
```

```
proc sgplot data=work.bank;  
    vbar y / datalabel;  
    xaxis label="Subscription Outcome (y)";  
    yaxis label="Count";  
run;  
  
title;  
footnote;
```